MULTIVARIABLE CALCULUS • UNIT 3.01

C Multivariable Optimization, Hessians & Lagrange Multipliers

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

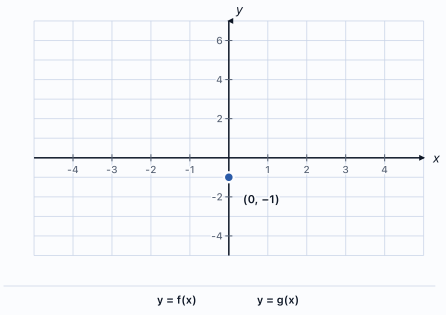
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 B	Q3 C	Q4 A	Q5 A
Q6 B	Q7 D	Q8 A	Q9 B	Q10 D
Q11 C	Q12 B	Q13 A	Q14 B	Q15 D
Q16 C	Q17 B	Q18 B	Q19 A	Q20 B
Q21 C	Q22 D	Q23 D	Q24 C	Q25 C

Multivariable Optimization, Hessians & Lagrange Multipliers Practice Exercises

1 MCQ

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.



A 10

B 13

C 26 ✓

D 52

WORKED SOLUTION & VERIFICATION

We are given $z = f(x, y) = x^2 + y^2$, with $x = 2t$ and $y = 3t$. We need to find $\frac{dz}{dt}$ using the Multivariable Chain Rule:
$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$.
First, find the partial derivatives of z with respect to x and y:
$\frac{\partial z}{\partial x} = 2x$
$\frac{\partial z}{\partial y} = 2y$
Next, find the derivatives of x and y with respect to t:
$\frac{dx}{dt} = 2$
$\frac{dy}{dt} = 3$
Now, substitute these into the chain rule formula:
$\frac{dz}{dt} = (2x)(2) + (2y)(3) = 4x + 6y$.
We are asked to find $\frac{dz}{dt}$ when $t = 1$. First, find the values of x and y when $t = 1$:
$x = 2(1) = 2$
$y = 3(1) = 3$
Finally, substitute these values of x and y into the expression for $\frac{dz}{dt}$:
$\left.\frac{dz}{dt}\right|_{t=1} = 4(2) + 6(3) = 8 + 18 = 26.$

2 MCQ

Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.

A 12	B **18** ✓
C 6	D 24

WORKED SOLUTION & VERIFICATION

To evaluate the double integral $\iint_R 2xy\, dA$ where $R = [0, 2] \times [0, 3]$, we can use Fubini's Theorem. We will set up the iterated integral by integrating with respect to y first, then x:

$$\int_0^2 \int_0^3 2xy\, dy\, dx$$

First, integrate with respect to y, treating x as a constant:

$$\int_0^3 2xy\, dy = [xy^2]_{y=0}^{y=3} = x(3^2) - x(0^2) = 9x$$

Now, integrate the result with respect to x:

$$\int_0^2 9x\, dx = [\frac{9x^2}{2}]_{x=0}^{x=2} = \frac{9(2^2)}{2} - \frac{9(0^2)}{2} = \frac{9(4)}{2} = \frac{36}{2} = 18$$

Therefore, the value of the double integral is 18.

3 MCQ

For $f(x, y) = 7x^2 + 4y^2$, find $f_y(-3, 1)$.

A -24	B 4
C **8** ✓	D -42

WORKED SOLUTION & VERIFICATION

Applying Partial Derivatives principles, $f_y = 8y$, so $f_y(-3, 1) = 8(1) = 8$. Therefore, the verified correct option is 8.

4 MCQ

Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.

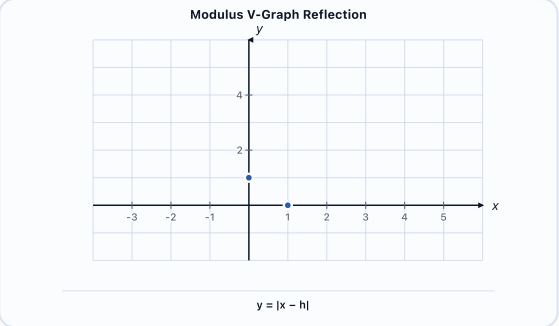
A **2** ✓	B 4
C 6	D 0

WORKED SOLUTION & VERIFICATION

First, we find the first partial derivative of $f(x, y)$ with respect to x: $f_x(x, y) = \frac{\partial}{\partial x}(x^2 + y^2) = 2x$.
Next, we find the second partial derivative of $f(x, y)$ with respect to x (which is f_{xx}): $f_{xx}(x, y) = \frac{\partial}{\partial x}(f_x(x, y)) = \frac{\partial}{\partial x}(2x) = 2$.
Finally, we evaluate $f_{xx}(x, y)$ at the point $(2, 3)$. Since $f_{xx}(x, y)$ is a constant, its value at any point is 2. Therefore, $f_{xx}(2, 3) = 2$.

5 MCQ

Let the linear scalar potential be $h(u, y) = 15u + 8y - 16$. Compute the squared gradient norm $\|\nabla h\|^2$.



- **A** 289 ✓
- **B** 293
- **C** 281
- **D** 304

WORKED SOLUTION & VERIFICATION

Since $\nabla h = \langle 15, 8 \rangle$, $\|\nabla h\|^2 = 15^2 + 8^2 = 289$.

6 MCQ

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.

- **A** $\sqrt{14}$
- **B** $2\sqrt{14}$ ✓
- **C** $3\sqrt{14}$
- **D** $4\sqrt{14}$

WORKED SOLUTION & VERIFICATION

The surface area of a surface $z = g(x, y)$ above a region R is given by the integral $\iint_R \sqrt{1 + (g_x)^2 + (g_y)^2}\, dA$. Here, $g(x, y) = 2x + 3y + 1$. The partial derivatives are $g_x = 2$ and $g_y = 3$. Therefore, $\sqrt{1 + (g_x)^2 + (g_y)^2} = \sqrt{1 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$. The integral becomes $\iint_R \sqrt{14}\, dA$, where $R = [0, 1] \times [0, 2]$. This evaluates to $\sqrt{14} \times \text{Area}(R)$. The area of the rectangle R is $(1 - 0) \times (2 - 0) = 2$. So, the surface area is $2\sqrt{14}$.

7 MCQ

Consider the vector field $\mathbf{u}(s, y, z) = \langle 22s,\ 7y,\ 7z \rangle$ on R^3. Compute the scalar divergence $\nabla \cdot \mathbf{u}$.

- **A** 58
- **B** 40
- **C** 29
- **D** 36 ✓

WORKED SOLUTION & VERIFICATION

The divergence is $22 + 7 + 7 = 36$.

8 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. Evaluate the vector field at the point $(2, 4)$.

- **A** $\langle 6, 8 \rangle$ ✓
- **B** $\langle 8, 6 \rangle$
- **C** $\langle 5, 6 \rangle$
- **D** $\langle 6, 5 \rangle$

WORKED SOLUTION & VERIFICATION

The vector field is given by $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. To evaluate the vector field at the point $(2, 4)$, we substitute $x = 2$ and $y = 4$ into the components of the vector field.

First component: $3x = 3(2) = 6$.

Second component: $2y = 2(4) = 8$.

Therefore, $\mathbf{F}(2, 4) = \langle 6, 8 \rangle$.

9 MCQ

Consider the function $f(x, y) = x^2 + y^2 - 4x - 6y + 13$. Find the coordinates of the critical point for this function.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	13	10	8	5

- **A** (-2, -3)
- **B** **(2, 3)** ✓
- **C** (4, 6)
- **D** (0, 0)

WORKED SOLUTION & VERIFICATION

To find the critical point(s) of a function of two variables, we need to find where the gradient is zero. The gradient is given by $\nabla f = \langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \rangle$.

First, we compute the partial derivative with respect to x:

$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2 + y^2 - 4x - 6y + 13) = 2x - 4$.

Next, we compute the partial derivative with respect to y:

$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2 + y^2 - 4x - 6y + 13) = 2y - 6$.

To find the critical point(s), we set both partial derivatives equal to zero and solve the resulting system of equations:

1) $2x - 4 = 0$

$2x = 4$

$x = 2$

2) $2y - 6 = 0$

$2y = 6$

$y = 3$

Therefore, the critical point is $(2, 3)$.

Alternative method: Completing the square.

We can rewrite the function by completing the square for the x and y terms:

$f(x, y) = (x^2 - 4x) + (y^2 - 6y) + 13$

$f(x, y) = (x^2 - 4x + 4) - 4 + (y^2 - 6y + 9) - 9 + 13$

$f(x, y) = (x - 2)^2 + (y - 3)^2$

This is a paraboloid that opens upwards. The minimum value occurs when $(x - 2)^2 = 0$ and $(y - 3)^2 = 0$, which means $x = 2$ and $y = 3$. This minimum point is the critical point of the function.

10 MCQ

Consider the vector field $\mathbf{h}(u, y, z) = \langle 4u,\ 10y,\ 5z \rangle$ on R^3. Compute the scalar divergence $\nabla \cdot \mathbf{h}$.

A 28	B 9
C 23	D **19** ✓

WORKED SOLUTION & VERIFICATION

The divergence is $4 + 10 + 5 = 19$.

11 MCQ

Consider the function $f(x, y) = x^3 + y^3 - 3axy$, where a is a positive constant. It is known that $f(x, y)$ has a critical point at (x_0, y_0) where $x_0 = 2$ and $y_0 = 2$. Find the value of a.

A 1	B 3
C **2** ✓	D 4

WORKED SOLUTION & VERIFICATION

To find the critical points of the function $f(x, y) = x^3 + y^3 - 3axy$, we first compute the first partial derivatives and set them to zero:

$f_x = 3x^2 - 3ay$

$f_y = 3y^2 - 3ax$

Setting these to zero, we get the system of equations:

1) $3x^2 - 3ay = 0$ => $x^2 = ay$

2) $3y^2 - 3ax = 0$ => $y^2 = ax$

We are given that $(x_0, y_0) = (2, 2)$ is a critical point. Substituting these values into the equations:

From equation 1: $2^2 = a(2)$ => $4 = 2a$ => $a = 2$.

From equation 2: $2^2 = a(2)$ => $4 = 2a$ => $a = 2$.

Both equations yield the same value for a, which is 2. This confirms that $a = 2$ is consistent with $(2, 2)$ being a critical point. The problem states a is a positive constant, and $a = 2$ satisfies this condition.

12 MCQ

Consider the vector field $\mathbf{f}(x, y, z) = \langle 12x,\ 20y,\ 11z \rangle$ on R^3. Determine the scalar divergence $\nabla \cdot \mathbf{f}$.

A 23	B **43** ✓
C 47	D 55

WORKED SOLUTION & VERIFICATION

The divergence is $12 + 20 + 11 = 43$.

13 MCQ

Let the multivariable function u be defined by $u(s, y) = 14s^2y + 24y^2$. Compute the partial derivative $\frac{\partial u}{\partial s}(6, 10)$.

- **A** **1680** ✓
- **B** 1656
- **C** 1684
- **D** 1694

WORKED SOLUTION & VERIFICATION

Differentiating with respect to s gives $28sy$, which equals 1680 at $(6, 10)$.

14 MCQ

Calculate $\nabla f(-2, -1)$ for $f(x, y) = 4x^2 + 8y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	4	8	12

- **A** (-8, -8)
- **B** **(-16, -16)** ✓
- **C** (16, 16)
- **D** (-16, -14)

WORKED SOLUTION & VERIFICATION

$\nabla f = (8x, 16y)$, so $\nabla f(-2, -1) = (-16, -16)$.

15 MCQ

Find $\nabla f(1, 3)$ for $f(x, y) = 7x^2 + 9y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	7	9	16

- **A** (54, 14)
- **B** (7, 27)
- **C** (-14, -54)
- **D** **(14, 54)** ✓

WORKED SOLUTION & VERIFICATION

$\nabla f = (14x, 18y)$, so $\nabla f(1, 3) = (14, 54)$.

16 MCQ

Establish $\nabla f(-3,-6)$ for $f(x,y) = 6x^2 + 4y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	6	4	10

A (-48, -36)

B (-18, -24)

C **(-36, -48)** ✓

D (36, 48)

WORKED SOLUTION & VERIFICATION

$\nabla f = (12x, 8y)$, so $\nabla f(-3,-6) = (-36,-48)$.

17 MCQ

Find $\nabla f(-1,4)$ for $f(x,y) = x^2 + 2y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	1	2	3

A (16, -2)

B **(-2, 16)** ✓

C (-1, 8)

D (2, -16)

WORKED SOLUTION & VERIFICATION

$\nabla f = (2x, 4y)$, so $\nabla f(-1,4) = (-2,16)$.

18 MCQ

Establish $\nabla f(-1,1)$ for $f(x,y) = 3x^2 + 9y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	3	9	12

A (18, -6)

B **(-6, 18)** ✓

C (-3, 9)

D (6, -18)

WORKED SOLUTION & VERIFICATION

$\nabla f = (6x, 18y)$, so $\nabla f(-1,1) = (-6,18)$.

19 MCQ

Find $\nabla f(-1,-2)$ for $f(x,y) = 4x^2 + y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	4	1	5

- **A (-8, -4)** ✓
- B (-4, -8)
- C (-4, -2)
- D (8, 4)

WORKED SOLUTION & VERIFICATION

$\nabla f = (8x, 2y)$, so $\nabla f(-1,-2) = (-8,-4)$.

20 MCQ

Find $\nabla f(-2,2)$ for $f(x,y) = 4x^2y + 4xy^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	0	0	8

- A (16, 16)
- **B (-16, -16)** ✓
- C (-15, -16)
- D (-32, -32)

WORKED SOLUTION & VERIFICATION

$f_x = 8xy + 4y^2, f_y = 4x^2 + 8xy$, so $\nabla f(-2,2) = (-16,-16)$.

21 MCQ

Deduce $\nabla f(4,-2)$ for $f(x,y) = 4x^2y + 5xy^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	0	0	9

- A (-64, -80)
- B (-16, -44)
- **C (-44, -16)** ✓
- D (44, 16)

WORKED SOLUTION & VERIFICATION

$f_x = 8xy + 5y^2, f_y = 4x^2 + 10xy$, so $\nabla f(4,-2) = (-44,-16)$.

22 MCQ

Compute $\nabla f(1,3)$ for $f(x,y) = 4x^2y + 5xy^2$.

- A (24, 30)
- B (34, 69)
- C (-69, -34)
- **D (69, 34)** ✓

WORKED SOLUTION & VERIFICATION

$f_x = 8xy + 5y^2, f_y = 4x^2 + 10xy$, so $\nabla f(1,3) = (69,34)$.

23 MCQ

Obtain $\nabla f(4, 3)$ for $f(x, y) = 5x^2y + 3xy^2$.

A (120, 72)

B (152, 147)

C (-147, -152)

D **(147, 152)** ✓

WORKED SOLUTION & VERIFICATION

$f_x = 10xy + 3y^2, f_y = 5x^2 + 6xy$, so $\nabla f(4, 3) = (147, 152)$.

24 MCQ

Determine $\nabla f(-3, -1)$ for $f(x, y) = 5x^2y + 2xy^2$.

A (30, 12)

B (57, 32)

C **(32, 57)** ✓

D (-32, -57)

WORKED SOLUTION & VERIFICATION

$f_x = 10xy + 2y^2, f_y = 5x^2 + 4xy$, so $\nabla f(-3, -1) = (32, 57)$.

25 MCQ

Deduce $\nabla f(2, -2)$ for $f(x, y) = x^2y + 5xy^2$.

A (-8, -40)

B (-36, 12)

C **(12, -36)** ✓

D (-12, 36)

WORKED SOLUTION & VERIFICATION

$f_x = 2xy + 5y^2, f_y = x^2 + 10xy$, so $\nabla f(2, -2) = (12, -36)$.